

MORGAN M. RYAN, M.A.

Grand Rapids, MI | 402-640-2933 | MorganRyan2014M@gmail.com | [LinkedIn](#) | [Portfolio](#)

GIS Developer with experience designing and supporting enterprise geospatial applications, automation workflows, and data pipelines using ArcGIS Enterprise, Python, JavaScript, and enterprise databases. Background includes utility GIS, enterprise systems integration, QA/QC automation, and scalable geospatial workflow development supporting operational and regulatory initiatives.

TECHNICAL SKILLS

GIS & Esri Platforms: ArcGIS Enterprise, ArcGIS Pro, ArcGIS Online (AGOL), ArcGIS Experience Builder, ArcGIS Dashboards, ArcGIS API for Python, ArcGIS JS API, ArcPy, Survey123, Field Maps, Web AppBuilder, Data Reviewer, Utility Network, VertiGIS Studio, ArcGIS Monitor

Programming & Development: Python, JavaScript, SQL, PowerShell, HTML/CSS, Arcade, Git, RESTful Services

Databases & Data Management: PostgreSQL/PostGIS, Oracle, Enterprise Geodatabases, SQL Server, SQLite, SDE Versioning, Spatial ETL, Data Modeling

Automation & Integration: Power Automate, REST APIs, ArcGIS REST API, Multiprocessing, ETL Workflows, QA/QC Automation, Geospatial Big Data Pipelines, Enterprise Workflow Automation, Service Administration, Configuration Management, Portal Automation

Data Science & Analysis: Pandas, Spatial Analysis, Remote Sensing, Hydrologic Modeling, GNSS Data Collection, Statistical Analysis, Big Data Analysis, Utility Network Modeling, Network Analysis

Cloud & Enterprise Systems: AWS Workspaces, Enterprise GIS Administration, Portal Administration, Map & Feature Services, Enterprise GIS Architecture, Identity/Access Management, High Availability Systems, Enterprise Service Management

PROFESSIONAL EXPERIENCE

PHILADELPHIA GAS WORKS — Geographic Information Team

Senior GIS Analyst / Developer | Philadelphia, PA | Oct 2023 – Present

- Design, develop, and maintain customer-facing enterprise GIS applications, dashboards, web maps, and geospatial workflows using ArcGIS Enterprise, Experience Builder, Python, and JavaScript.
- Develop scalable geospatial services, automation pipelines, and spatial workflows supporting operational, permitting, compliance, and field applications.
- Build automated GIS data processing, QA/QC validation, and reporting workflows using Python, ArcPy, SQL, PostgreSQL, Oracle, and enterprise geodatabases.
- Administer and support ArcGIS Enterprise environments, including portal configuration, map and feature services, enterprise data accessibility, and application performance optimization.
- Lead modernization and migration efforts for legacy GIS applications and Data Reviewer workflows into modern ArcGIS Enterprise and Experience Builder architectures.
- Engineer large-scale geospatial automation and validation workflows using ArcPy, multiprocessing, Data Reviewer, and enterprise geodatabases.
- Collaborate with technical and business stakeholders to define GIS requirements and develop scalable geospatial solutions.

GEOGRAPHIC CONSULTING GROUP INTERNATIONAL

Solutions Engineer | Alexandria, VA | Jul 2023 – Oct 2023

- Developed and configured Electric Utility Network models for U.S. military installations aligned with Esri Utility Network standards and enterprise geospatial architecture requirements.

- Designed and delivered technical presentations and solution demonstrations supporting Utility Network adoption across domestic and international military installations.
- Built Python automation tools and validation workflows to assess GEOFidelis data model readiness for Utility Network implementation and migration planning.
- Led technical training sessions covering Utility Network architecture, spatial analysis workflows, and geospatial application development for analysts and project stakeholders.
- Collaborated with cross-functional teams to support enterprise GIS modernization initiatives and utility network implementation strategies.

OFFICE OF THE CHIEF INFORMATION OFFICER — Geographic Information Office

Senior IT Applications Developer | Lincoln, NE | Jul 2022 – Jul 2023

- Developed and automated enterprise GIS data processing and administration workflows using Python, JavaScript, PowerShell, and SQL.
- Utilized Pandas, ArcPy, and ArcGIS API for Python to streamline large-scale spatial analysis, data management, and geospatial reporting processes.
- Designed and maintained GIS portal administration applications and automation tools supporting enterprise GIS operations and statewide agency users.
- Managed SQL databases and enterprise geospatial data workflows supporting operational analysis and business decision-making.
- Provided technical GIS application support and enterprise geospatial solutions for multiple Nebraska state agencies.

NEBRASKA DEPARTMENT OF NATURAL RESOURCES — Floodplain Division

Environmental Specialist I | Lincoln, NE | Aug 2021 – Jul 2022

- Conducted watershed-scale hydrologic and floodplain modeling supporting FEMA submissions and Base Flood Elevation analyses.
- Created technical documentation and reports detailing flood analysis methodologies and GIS workflows.
- Developed automated geospatial workflows supporting hydrologic analysis and environmental review processes.

EDUCATION

UNIVERSITY OF NEBRASKA—LINCOLN

Master of Arts in Geography, 2022 | GPA: 3.93

Focus Areas: GISystems, Web Cartography, Remote Sensing, Spatial Access, Health Geography

Bachelor of Science in Education, 2019

Endorsement: Social Sciences | Minors: Geography, History

RESEARCH EXPERIENCE

Reexamining the Desert: A Study of Place-Based Food Insecurity — Published Master's Thesis

- Expanded USDA food desert identification methodologies using place-specific variables.
- Developed spatial models analyzing food insecurity and accessibility.
- Conducted mixed-method geospatial analysis and field research across 52 grocery stores in Lincoln, Nebraska.

TEACHING & LEADERSHIP EXPERIENCE

UNIVERSITY OF NEBRASKA—LINCOLN — Graduate Teaching Assistant | Aug 2019 – Spring 2021

- Created GIS labs for a published Physical Geography Lab Manual.
- Adapted GIS coursework for ArcGIS Pro and QGIS environments.
- Taught undergraduate and graduate GIS labs covering ArcGIS Pro, QGIS, Leaflet, ArcGIS API for JavaScript, D3.js, and advanced spatial analysis techniques.
- Developed and led a three-hour conference seminar focused on web mapping with QGIS and Leaflet.